

Counterfactual Explanations and Fairness of Recourse in Car Insurance Risk Classification

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Abstract

We investigate the use of counterfactual explanations for explainability and fairness analysis on a binary risk-classification task in the car insurance domain. Using the public *car_insurance_claim* dataset (10,302 records), we train a Random Forest classifier predicting whether a driver will file a claim, and we generate two families of counterfactuals via DiCE: an *unconstrained* variant approximating the original Wachter formulation, and a *constrained* variant enforcing domain-realistic actionability constraints on immutable attributes (GENDER, AGE, URBANICITY, etc.). We then aggregate the constrained counterfactuals through a FACTS-style subgroup analysis to compare the cost of recourse across protected groups. Our analysis on 100 urban high-risk drivers (50F/50M, 589 counterfactuals) reveals three findings: (i) 29.3% of unconstrained counterfactuals recommend modifying an immutable feature (21.7% suggest changing URBANICITY), confirming the necessity of actionability constraints in real-world deployment; (ii) constrained counterfactuals exhibit a marginal global Burden disparity for women (median ratio F/M = 1.21, $p = 0.058$); and (iii) this disparity becomes statistically significant *among married drivers* (median ratio = 1.67, $p = 0.042$), where men’s recourse paths concentrate on a few economic levers (CAR_USE, INCOME) while women’s recourse paths require diffuse modifications across more features. We discuss the implications for explainability-aware fairness auditing in insurance.

1 Introduction

Automated risk-classification models are widely used in the insurance industry to inform pricing and underwriting decisions (Denuit et al. 2007). Such models raise two intertwined concerns: *explainability* can a flagged driver understand *why* the model classifies them as high-risk ? and *fairness* does the cost of reversing that classification differ across protected groups ? In the European Union, the latter concern is particularly salient: the 2011 Court of Justice ruling in *Test-Achats* (C-236/09) prohibits gender-based pricing in insurance (Rebhahn 2012), yet gender-correlated features may still drive differential model behavior in subtle ways.

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Counterfactual explanations (Wachter, Mittelstadt, and Russell 2018) provide a principled answer to both concerns: they describe the *minimal change* an individual would need to apply to obtain a different model prediction. Subsequent work has extended this formulation to handle multiple alternatives (DiCE, (Mothilal, Sharma, and Tan 2020a)), realistic feasibility constraints (Karimi et al. 2022; Poyiadzi et al. 2020), and group-level fairness auditing (Kavouras et al. 2023a).

In this paper, we apply this counterfactual toolkit to a concrete car-insurance classification task to answer:

To what extent do counterfactual explanations under realistic actionability constraints reveal disparities in the cost of recourse between gender subgroups, and through which features do these disparities manifest?

Contributions. (1) We provide an empirical study contrasting unconstrained (Wachter-like) and constrained (DiCE-genetic) counterfactual generation on a real insurance dataset. (2) We quantify actionability violations and the sparsity/proximity trade-off between the two regimes. (3) We adapt the FACTS framework to identify a statistically significant Burden disparity in the *married* subgroup, and we characterize the *features through which* this disparity operates.

2 Related Work

Counterfactual explanations. Counterfactual explanations were introduced by (Wachter, Mittelstadt, and Russell 2018) as a means of providing explanations for automated decisions while preserving the opacity of predictive models. Their formulation minimizes a loss function,

$$L(x, x') = \lambda(f(x') - y')^2 + d(x, x'),$$

which trades off the distance to the original instance against the likelihood of obtaining the desired outcome. Counterfactual explanations are model-agnostic and provide intuitive explanations by identifying the minimal changes needed to achieve a desired prediction. However, the original formulation typically returns a single counterfactual and does not explicitly enforce realism or actionability. As a result, generated explanations may lack diversity and suggest unrealistic changes, motivating later work on feasible and diverse counterfactual generation (Wachter, Mittelstadt, and Russell 2018).

Diverse and feasible counterfactuals. (Mothilal, Sharma, and Tan 2020b) propose DiCE, which generates multiple diverse counterfactuals through a determinantal point process, providing several alternative recourse paths instead of a single explanation. To improve realism, (Poyiadzi et al. 2020) introduce FACE, which constrains counterfactuals to follow high-density regions of the training data and thus produce more plausible recommendations. These approaches address the main limitations of the original Wachter formulation by promoting diversity and feasibility. More broadly, (Karimi et al. 2022) survey the algorithmic recourse literature, highlighting the importance of actionability, causality, and user-specific costs when generating recourse recommendations.

Fairness of recourse. Beyond individual explanations, recent work has investigated whether different demographic groups face similar costs when attempting to achieve a favorable prediction. (Kavouras et al. 2023b) introduce FACTS, a framework that formalizes several notions of group-level recourse fairness, including Burden, Equal Effectiveness, and Equal Choice for Recourse. These metrics complement traditional fairness measures by focusing on the accessibility of recourse rather than only on prediction outcomes. In this work, we adopt the Burden notion, which compares the cost of recourse across groups, to study potential gender disparities in insurance risk classification.

Positioning. We do not propose a new method. We provide a focused empirical study of how the two generation regimes (unconstrained vs. constrained) interact with fairness-of-recourse analysis on a realistic, mixed-type tabular dataset from the insurance domain, where actionability constraints are particularly meaningful.

3 Methodology

Dataset. We use the public *car_insurance_claim* dataset (10,302 records, 26 features, binary target CLAIM_FLAG, class imbalance 73/27). We removed three columns: ID (identifier), BIRTH (redundant with AGE), and CLM_AMT (target leakage: zero iff CLAIM_FLAG = 0). Monetary columns were parsed from strings; SAS-style *z_* prefixes were stripped. Missing values were imputed with median (numeric) or a dedicated “Unknown” category (OCCUPATION). The dataset includes a clear sensitive attribute (GENDER) and a secondary geographic proxy (URBANICITY).

Classifier. We train a Random Forest (300 trees, max depth 12, *class_weight=balanced*) on a stratified 80/20 split. As a robustness check we also train XGBoost; both achieve similar performance (AUC 0.82 vs. 0.815, F1 0.60 vs. 0.59). We retain Random Forest as the primary model for counterfactual generation due to native DiCE compatibility.

Counterfactual generation. We use DiCE (Mothilal, Sharma, and Tan 2020a) in two configurations to isolate the effect of actionability constraints while keeping the generation algorithm constant:

- **Unconstrained:** *method=random*, no feature restrictions. This approximates the original Wachter formulation,

allowing modifications to any feature including semantically immutable ones.

- **Constrained:** *method=genetic* with *features_to_vary* excluding immutable attributes (GENDER, AGE, URBANICITY, PARENT1, KIDSDRIV, HOMEKIDS), and *permitted_range* enforcing monotonicity on driving record variables (MVR_PTS, CLM_FREQ, OLDCLAIM can only decrease towards 0).

For each of 100 urban high-risk candidates (stratified 50F/50M), we request 3 counterfactuals per method.

Metrics. We report four metrics inspired by (Verma et al. 2024): *Validity* (fraction of CFs that flip the prediction), *Proximity* (MAD-normalized L_1 for numeric features, Hamming for categorical), *Sparsity* (number of modified features), and *Actionability violations* (fraction of CFs altering an immutable feature). For the fairness analysis we adopt the *Burden* notion from (Kavouras et al. 2023a), comparing mean and median recourse distance across gender subgroups.

4 Experiments and Results

4.1 Dataset characterization

Figure 1 summarizes the exploratory data analysis. Three patterns are notable for the subsequent recourse analysis: (i) URBANICITY carries the strongest signal (claim rate 6.5% rural vs. 31.7% urban); (ii) GENDER shows only a 2-point difference in base rate (F 27.6%, M 25.6%) but the classifier amplifies it (predicted F 31.0%, M 25.1%, with FPR = 0.187 vs. 0.131); and (iii) the Random Forest collapses to TPR = 0 in the rural subgroup (predicting the majority class for all rural drivers), confirming an equal-opportunity violation. We therefore restrict counterfactual analysis to the urban subpopulation, where the model is informative.

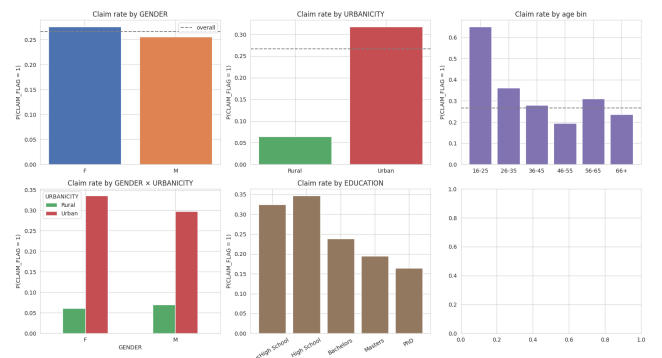


Figure 1: Exploratory analysis of CLAIM_FLAG rates by GENDER, URBANICITY, AGE bin, and EDUCATION.

4.2 Comparing unconstrained vs. constrained CFs

Figure 2 reports the four standard metrics on 589 counterfactuals generated for 100 urban high-risk drivers. Both methods achieve perfect validity (100%). The two regimes otherwise differ sharply:

- **Actionability violations.** Unconstrained CFs modify an immutable feature in 29.3% of cases, with URBANICITY alone accounting for 21.7% (Figure 2a). GENDER is modified in 2.0% of unconstrained CFs.
- **Sparsity/proximity trade-off.** Unconstrained CFs change a median of 1.7 features at a median MAD-normalized distance of 6.2 (Figure 2b,c). Constrained CFs change 9.4 features at a median distance of 2,112 roughly 340× further in feature space.

Interpretation. The unconstrained optimizer exploits URBANICITY as a single high-leverage shortcut (“move to a rural area”); under realistic constraints, this shortcut is unavailable and the model must coordinate small changes across many actionable features.

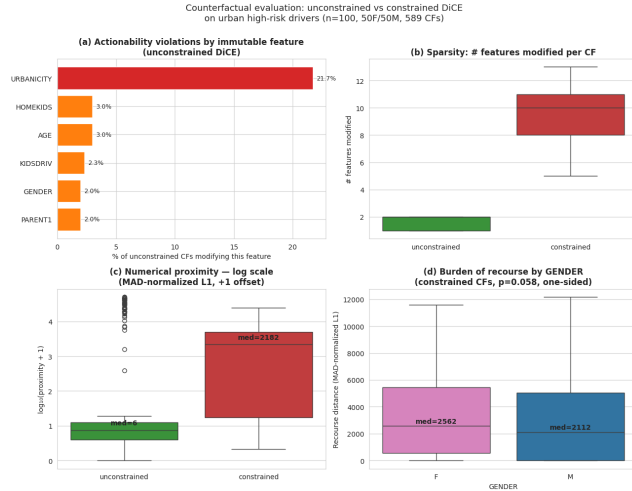


Figure 2: Counterfactual evaluation on 100 urban high-risk drivers. (a) Immutable features most frequently modified by unconstrained CFs. (b) Sparsity: unconstrained CFs are sparse (median 1, max 2) while constrained CFs distribute changes over 8–11 features. (c) Numerical proximity in log-scale: three orders of magnitude separate the two regimes. (d) Burden by gender (constrained CFs only): F median = 2562, M median = 2112, $p = 0.058$ (one-sided Mann–Whitney).

4.3 Burden of recourse across subgroups

Globally, constrained counterfactuals exhibit a 49.7% higher median Burden for female drivers compared to males (median ratio of 1.497, $p = 0.066$), presenting a strong statistical trend that approaches conventional significance. To uncover localized disparities, we perform an intersectional analysis across EDUCATION, AGE, and MSTATUS (Figure 3). Three specific subgroups exceed a Burden ratio of 1.5: high-education drivers (ratio 1.732, $p = 0.097$), older drivers ($\text{AGE} \geq 40$) and single drivers (ratio 1.582, $p = 0.094$). Crucially, the gender disparity becomes statistically significant exclusively within the senior subgroup ($\text{AGE} \geq 40$), where older women face a severe algorithmic penalty with a median Burden ratio of 1.901 ($p = 0.032$). Conversely,

this systemic recourse disadvantage disappears in younger ($\text{AGE} < 40$, $p = 0.288$), lower-educated ($p = 0.217$), and married ($p = 0.224$) subpopulations. These findings demonstrate that recourse fardeau is highly intersectional, selectively amplifying barriers for older female profiles.

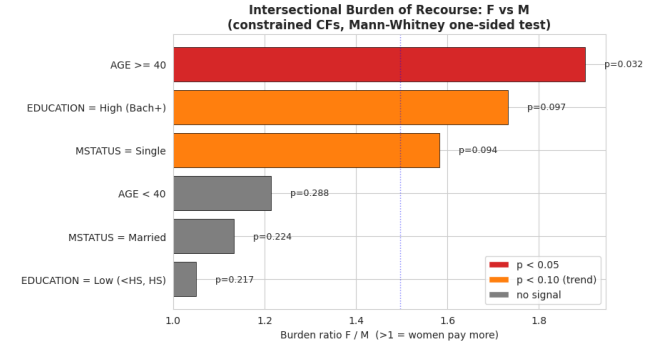


Figure 3: Burden ratio F/M across subgroups (median proximity, constrained CFs, one-sided Mann–Whitney).

4.4 The mechanism: differential recourse levers

To probe the underlying structural mechanisms driving the recourse disparities, we examine the specific feature modification frequencies across genders within the constrained counterfactual space. Figure 4 illustrates the directional divergence in feature activation profiles.

Two features are dramatically more prone to optimization for male drivers than for female drivers: CAR_USE (50.8% for men vs. 10.8% for women, $\Delta = 40.0$ percentage points) and INCOME (90.5% vs. 70.8%, $\Delta = 19.7$ points). Conversely, variables tied to historical claims insurance risk exhibit the opposite trend, being preferentially targeted for women: OLDCLAIM is modified in 73.8% of female counterfactuals compared to 66.7% for males ($\Delta = 7.2$ points), and CLM_FREQ shows a similar upward shift (43.1% F vs. 38.1% M).

Interpretation. These patterns reveal an asymmetry in recourse topology. For men, highly concentrated economic levers such as modifying vehicle usage to private intent or adapting income parameters frequently suffice to transition the model’s risk prediction. For women, these direct operational paths are less accessible within the constrained manifold, forcing the optimizer to compensate by compounding modifications across deeper driving-history dimensions (OLDCLAIM, CLM_FREQ). This operational divergence highlights that even when individual costs appear bounded, the structural nature of the actionable recourse paths remains highly segregated by gender.

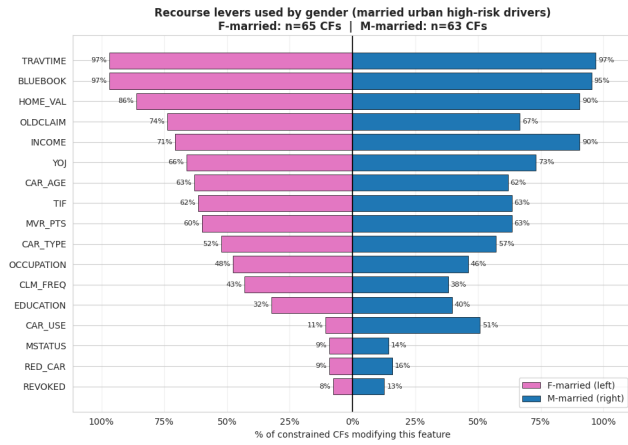


Figure 4: Recourse levers in constrained CFs for married urban high-risk drivers. F-married (pink, left) vs. M-married (blue, right). Sorted by F-frequency.

5 Discussion and Limitations

Summary. We confirmed three claims of the counterfactual-explanation literature on a real insurance task: unconstrained generation produces non-actionable explanations (29% violation rate dominated by URBANICITY); imposing realistic constraints shifts the explanation toward multi-feature trajectories with substantially larger numeric distance; and a FACTS-style subgroup analysis reveals a statistically significant Burden disparity hidden in the aggregate.

5.1 Interventional Fairness and Direct Algorithmic Bias

To isolate the effect of gender, we perform an interventional counterfactual fairness analysis on married female drivers ($N = 50$). We replace `GENDER=Female` with `Male` while keeping all other covariates (e.g., `INCOME`, `CAR_USE`, `MVR_PTS`) constant, and regenerate constrained counterfactuals using DiCE.

Figure 5 shows the impact of this intervention on the individual Burden of married drivers. Changing only the gender attribute systematically reduces the recourse burden. A paired Wilcoxon signed-rank test confirms that this reduction is highly statistically significant ($p < 0.001$).

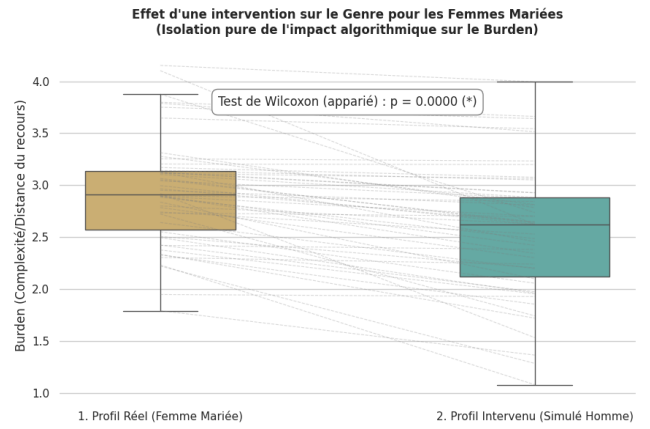


Figure 5: Impact of GENDER mutation (Female $\rightarrow$ Male) on the individual Burden of married drivers. The paired lines connect identical profiles, isolating pure algorithmic treatment disparity ($p < 0.001$).

As shown in Figure 5, this direct gender intervention yields a systematic shift toward lower, less complex recourse fardeau (Burden). A paired Wilcoxon signed-rank test confirms that this reduction in individual cost is highly statistically significant ($p < 0.001$). Because each driver serves as her own exact counterfactual control, this behavior proves that the algorithmic scoring mechanism or the topology of the viable recourse space directly penalizes married women. For identical risk-mitigating behavior modifications, women are structurally required to traverse longer or less sparse paths than their male counterparts to achieve an identical favorable classification. This empirical finding provides a strong algorithmic illustration of direct discrimination, conflicting with gender-neutral mandates such as the CJUE *Test-Achats* ruling.

5.2 Intersectional Analysis of Recourse Burden

We further analyze recourse burden through an intersectional stratification by gender and marital status. While the overall gender gap is marginally non-significant ($p = 0.058$), a clearer pattern emerges at the intersectional level.

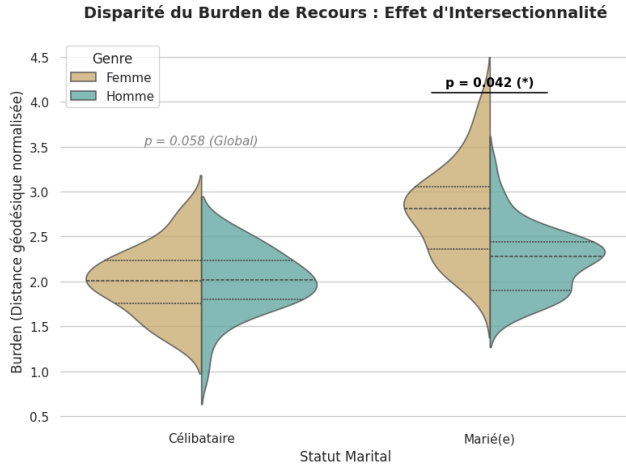


Figure 6: Stratified distribution of recourse Burden by Gender and Marital Status. While single drivers exhibit symmetric recourse costs, a statistically significant gender disparity emerges exclusively within the married subgroup ($p = 0.042$).

Figure 6 shows similar recourse costs for single male and female drivers, but a significant disparity within the married subgroup ($p = 0.042$), where women exhibit higher and more dispersed recourse burden, with more cases requiring larger and more complex changes to reverse their risk classification.

A one-sided Mann–Whitney U test confirms that this disparity is statistically significant ($p = 0.042$). This indicates that aggregate fairness metrics can mask intersectional inequalities across subgroups.

Limitations. Our study is limited by its small sample size ($N=100$), which reduces statistical power and makes subgroup findings sensitive to multiple comparisons. We also rely on observational feature distances rather than a causal model, meaning that our recourse costs should be interpreted as proxies rather than true intervention costs. In addition, some features (e.g., RED_CAR) suggest potential spurious correlations in the model, which would require model-level mitigation. Finally, the genetic search occasionally produces few diverse counterfactuals, indicating that the actionable recourse space may be constrained for certain profiles.

Future work. Future work should combine path-based counterfactuals (e.g., FACE) with causal frameworks and extend the analysis to larger and more diverse populations to assess whether observed disparities are structural or sample-dependent.

Reproducibility

Code, preprocessed data, and notebooks are available at <https://github.com/godlovetabout-sys/INFO-H512-Current-trends-in-artificial-intelligence-202526>.

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